

DATA VISUALISATION 2

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Changes in Migration in Australian Communities

Domain, Audience, and Purpose.

With an emphasis on cultural diversity and settlement across Statistical Area Level 3 (SA3) regions, this project investigates how migration transforms Australian communities. Planners, policy analysts, local councils, community organizations, and well-informed citizens who require an evidence-based perspective on the locations of overseas-born residents, the diversity of those communities, and the relationship between recent migration pulses and socioeconomic conditions are the target audience.

Data (what, sources, why appropriate).

- **ABS Census SA3 metrics (2011, 2016, 2021):** counts of total population, overseas-born, Simpson Diversity Index, and 2017–2021 new-arrival indicators compiled to SA3. Author: Australian Bureau of Statistics (ABS). Source: ABS Census DataPacks / TableBuilder extracts. These provide consistent small-area measures over time.
- **SEIFA 2021 (IRSAD):** Socio-Economic Indexes for Areas—Index of Relative Socio-economic Advantage and Disadvantage at SA2, 2021 release. Author: ABS; latest release on the SEIFA product page. We aggregate SA2 IRSAD to SA3 (using the first five digits of SA2 → SA3) to compare socio-economic position with migrant mix.
- **Migration Australia (Net Overseas Migration, calendar years 2004–2020):** national annual NOM series from ABS. This gives macro context for peaks/dips that drive local settlement patterns.

Links:

Census DataPacks: <https://www.abs.gov.au/census/find-census-data/datapacks>

SEIFA: <https://www.abs.gov.au/statistics/people/people-and-communities/socio-economic-indexes-areas-seifa-australia/latest-release>

Migration Australia:

<https://www.abs.gov.au/statistics/people/population/migration-australia>

Getting ready and joining.

I retained SA3_NAME21 for labels and standardized keys to SA3_CODE21. Substring (SA2_CODE, 0–5) was used to group SEIFA SA2 records to SA3. Only 2021's new-arrival rate and share are retained. ABS SA3 2021 TopoJSON is used for boundaries. With the exception of the "ZZZZZ" non-spatial code, all numeric fields were forced to numbers and their coverage was examined.

Design (color, layout, interaction, and idioms).

- A. The SA3 choropleth exhibits spatial variation in either Simpson Diversity (0–1), Overseas-born (%), or New Arrivals (2017–2021). Area-based intensity is effectively conveyed by choropleths; clipping is avoided with a Natural Earth projection. The scale is anchored by a legend, and full SA3 details are revealed through tooltips. Quick comparison is made possible by year/metric dropdowns (2011/2016/2021).

- B. With a labeled year axis and a readable grid, the Net Overseas Migration line (2004–2020) offers national context. The macro pulse (pre-GFC rise, pre-COVID peak, 2020 drop) is highlighted by a single color and a smoothed line.

- C. SEIFA vs. Migration (SA2) effectively summarizes the distribution rather than isolated points by using box-and-jitter by IRSAD decile to display the distribution and outliers of the overseas-born share across socioeconomic tiers.

D. SEIFA vs. migration (SA3) highlights different pockets in mid-to-higher advantage areas by using a dashed regression line and a red-dot scatter (mean IRSAD per SA3 vs. overseas-born %) to display the overall slope, which is somewhat negative.

- E. Diversity vs. Population (SA3) determines whether only the largest SA3s are diverse (they aren't) by substituting a red mean trend line across population ranges and a binned boxplot summary for a noisy scatter.

Why these decisions were made.

Map-first exploration (where), national context (when), and socioeconomic relationships (why/so what) are the next steps in the combination. The layout pairs each graphic with a brief "how to read/what it shows" note to assist non-expert audiences, and the colors follow intuitive schemes (purples/oranges for new arrivals, greens for diversity, and blues for shares).